Newton's Method: Short Report

Findings on Newton's method from the Project 4 nonlinear equation solver experiments.

Summary

Newton's method converges very quickly when it works, with quadratic convergence and few iterations. Its success depends on the starting point and on a nonzero derivative, and the solver reports each failure with a specific message.

1. Speed and Convergence

On $f(x) = x^2 - 2$ (exact root $\sqrt{2} = 1.414213562373\!10$), with $x_0 = 1$, $f'(x) = 2x$ and tolerance 10^{-10}, Newton converged in 4 iterations with a final error of 1.60×10^{-12}. The error roughly squares at each step, falling from 2.45×10^{-3} to 2.12×10^{-6} to 1.6×10^{-12}. This is quadratic convergence.

2. Effect of the Starting Point

Newton was applied to $f(x) = \arctan(x)$, whose only root is $x = 0$.

x_0	Converged	Iterations	Outcome
0.5	Yes	3	Root found
1.0	Yes	5	Root found
1.3	Yes	6	Root found
1.4	No	8	Diverging ($\lvert x\rvert > 10^8$)
2.0	No	4	Diverging ($\lvert x\rvert > 10^8$)

For large $|x|$, the derivative $1/(1 + x^2)$ is small, so a Newton step overshoots to the other side of the root. Beyond a critical value near $|x_0| \approx 1.392$, the iterates alternate in sign and grow. Starting closer to that value also costs more iterations.

3. Failure Cases

- **2-cycle:** on $x^3 - 2x + 2$ from $x_0 = 0$, the iterates cycle 0, 1, 0, 1, … and never approach the root. Only the iteration limit (20) detects this.

- **Better start:** the same function from $x_0 = -1$ converges to −1.769292354238631 in 8 iterations, so the failure comes from the initial guess.
- **Zero derivative:** on $x^2 - 1$ from $x_0 = 0$, $f'(0) = 0$. The solver stops before dividing by zero and reports the problem.

Conclusion

Newton's method is fast and accurate near the root, but it is not guaranteed to converge. It can diverge, fall into a cycle, or hit a zero derivative. Choose the starting point carefully, or pair it with a safer method such as bisection to find a good start.